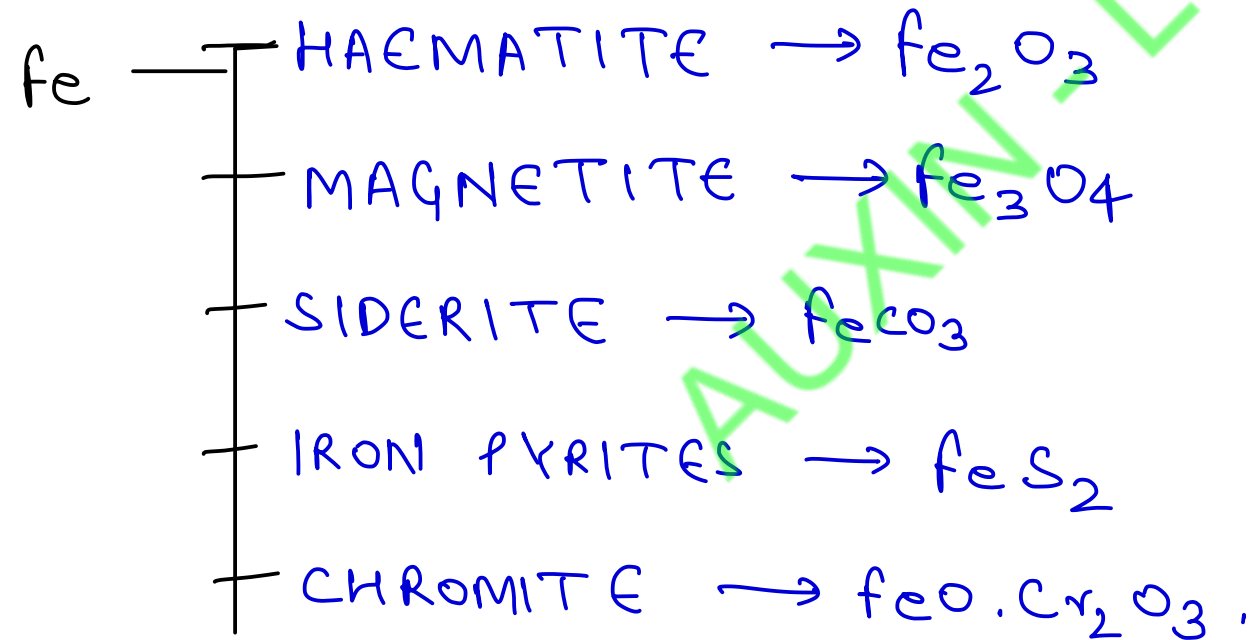
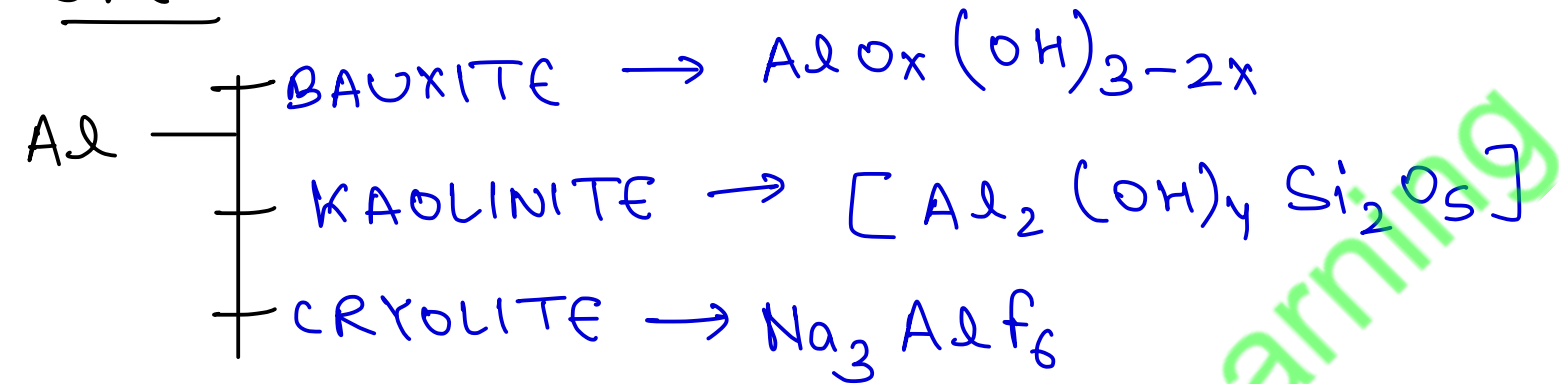
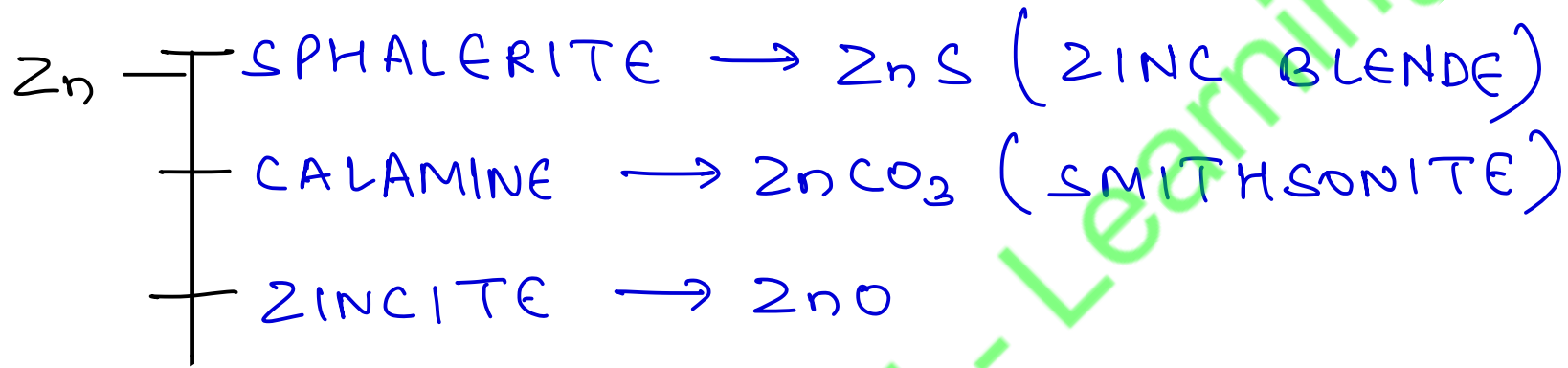
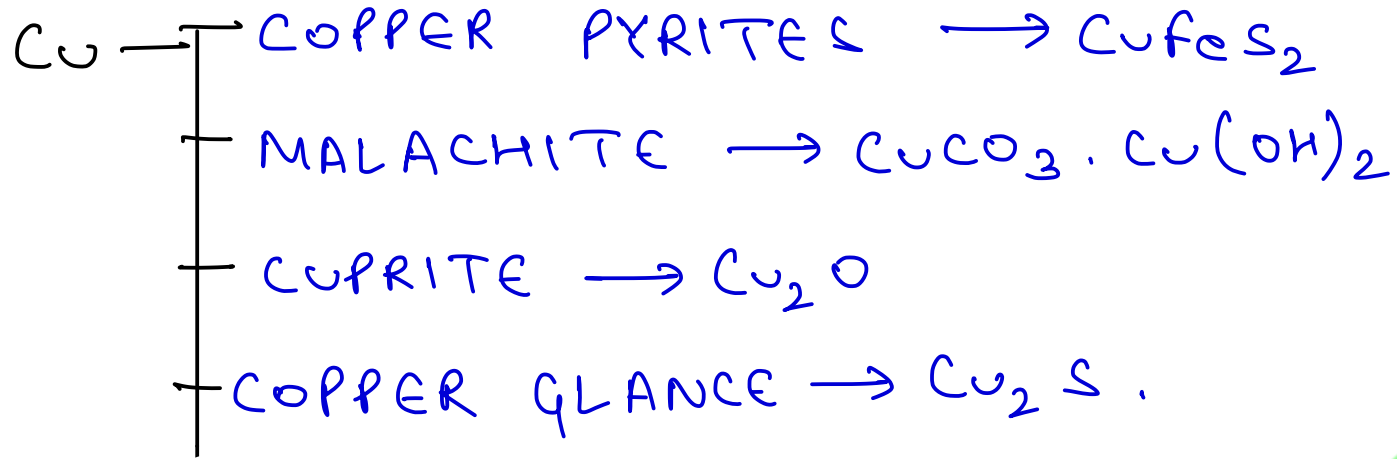


METALLURGY $\Rightarrow$

ORES $\rightarrow$





MOST ABUNDANT METAL IN EARTH'S CRUST $\rightarrow$

$\text{Al}(\text{III})$ followed BY $\text{IRON}(\text{IV})$
 $\downarrow$
 IN TERMS OF ELEMENT.

① CONCENTRATION of ORE



② ISOLATION of METAL from ITS CONCENTRATED ORE



③ PURIFICATION of METAL.

1 → CONCENTRATION of ORE →

— HYDRAULIC WASHING (LEVIGATION) → GRAVITY SEPARATION

— MAGNETIC SEPARATION → DIFFERENCE IN MAGNETIC PROP.

— FROATH-FLOATATION METHOD → for SULPHIDE ORES

— LEACHING → IF ORE IS SOLUBLE IN SUITABLE SOLVENT

— COLLECTOR → PINE OILS, FATTY ACIDS, XANTHATES

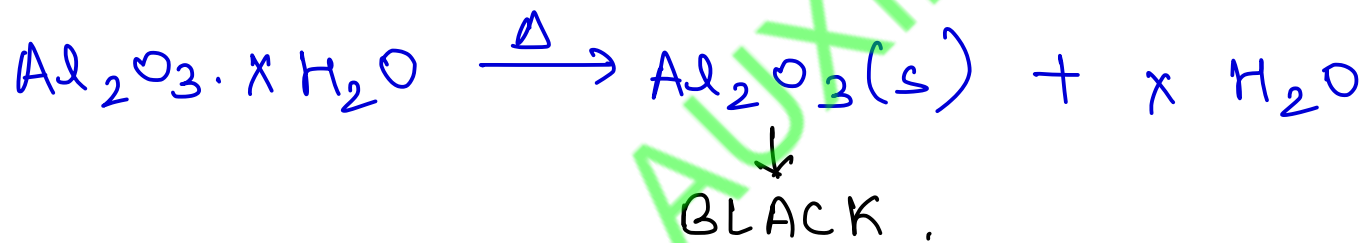
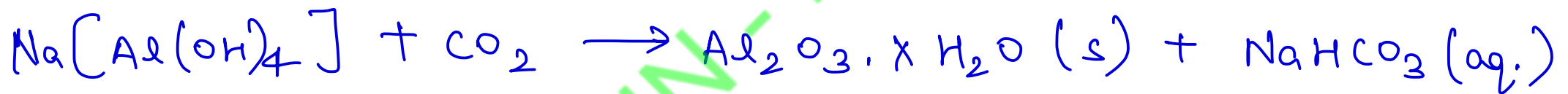
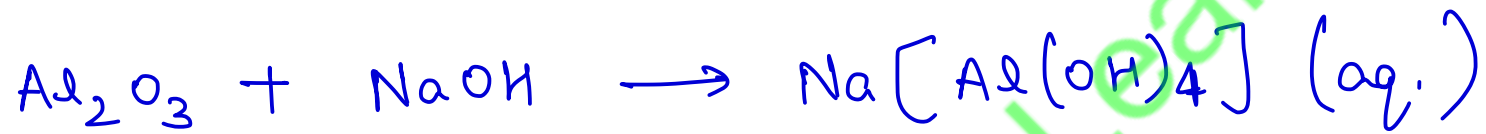
FROATH-FLOAT. — STABILIZER → ANILINE, CRESOL.

DEPRESSANTS $\rightarrow$ NaCN $\left\{ \begin{array}{l} \text{ZnS } \times \rightarrow \text{NOT ALLOWED BY NaCN} \\ \text{PbS } \checkmark \rightarrow \text{COME TO FROTH.} \end{array} \right.$

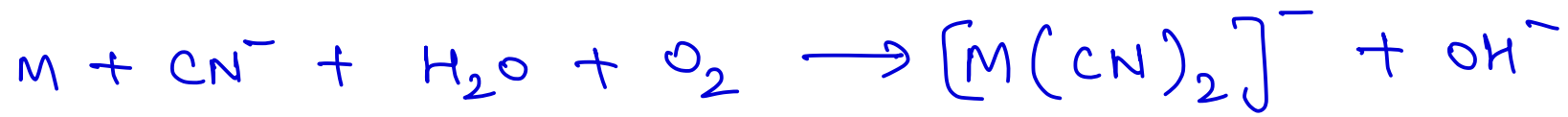
$\Rightarrow$ LEACHING $\Rightarrow$

① LEACHING OF ALUMINA FROM BAUXITE.

IMPURITIES $\rightarrow$ SiO₂, IRON OXIDES, TITANIUM OXIDES (TiO₂).



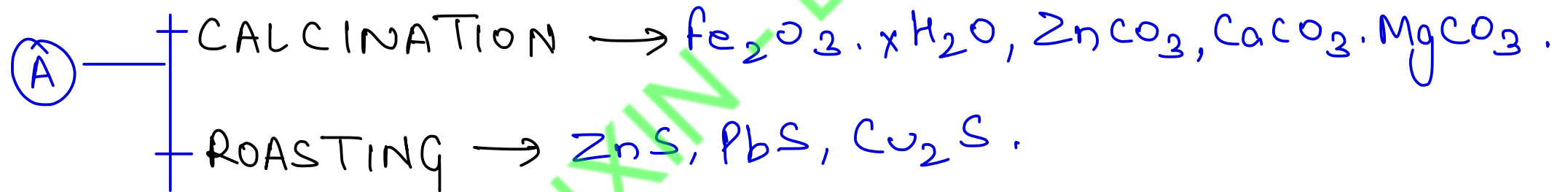
② IN METALLURGY OF SILVER AND GOLD $\rightarrow$ LEACHED WITH NaCN OR KCN IN PRESENCE OF O₂.



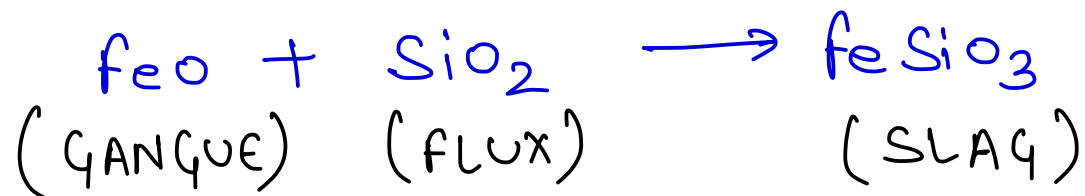
2 $\rightarrow$ EXTRACTION OF CRUDE METAL FROM ITS CONC. ORE $\rightarrow$

(A) CONVERSION TO OXIDE

(B) OXIDE TO METAL



REVERBATORY FURNACE $\rightarrow$ SULPHIDE ORES OF Cu ARE HEATED.



⇒ Cu PRODUCED IN FORM OF COPPER MATTE (Cu_2S & FeS)

ELLINGHAM DIAGRAM ⇒



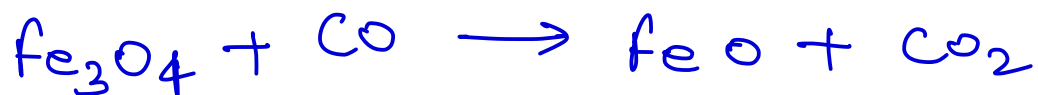
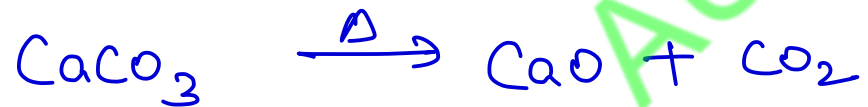
POINT TO REMEMBER →

① THE BELOW LINE METAL WILL REDUCE UPPER LINE OXIDE.

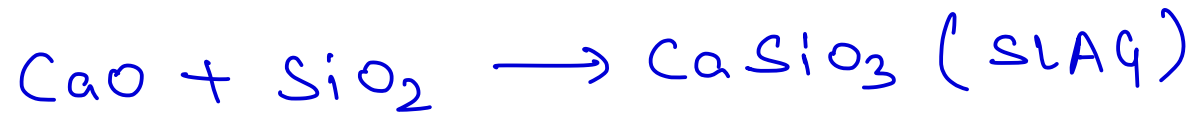
② Mg CAN REDUCE Al_2O_3 BELOW 1350°C AND ABOVE 1350°C MgO IS REDUCED BY Al.

③ → EXTRACTION OF IRON FROM ITS OXIDES → BLAST FURNACE

⇒ AT ABOUT 500-800 K →



⇒ AT ABOUT 900 - 1500 K →



BURNING OF COKE (ZONE OF COMBUSTION) →



ABOVE 1073 K → C CAN REDUCE FeO.

ZONE OF REDUCTION



ZONE OF SLAG FORMATION



ZONE OF HEAT ABSORPTION



ZONE OF COMBUSTION

IRON OBTAINED FROM BLAST FURNACE → PIG IRON (4% C)

IMPURITIES → S, P, Si, Mn

↓
CAST IRON (3% C)

↓
EXTREMELY HARD AND BRITTLE.

PUREST FORM OF IRON → WROUGHT IRON OR MALEABLE IRON.

PREPARED BY OXIDATIVE REFINING IN REVERBATORY FURN.



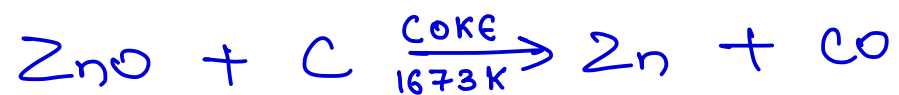
EXTRACTION OF Cu FROM CUPROUS (I) OXIDE →



DUE TO EVOLUTION OF SO_2 , IT GIVES BUSTER APPEAR-
ANCE.

↑
BLISTER COPPER

EXTRACTION OF Zn →



↳ DISTILLED OFF AND COLLECTED BY RAPID CHILLING.

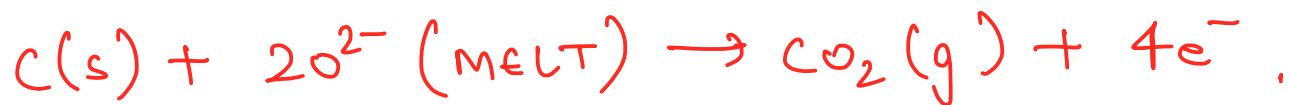
⇒ METALLURGY OF ALUMINIUM → HALL HEROULT PROCESS

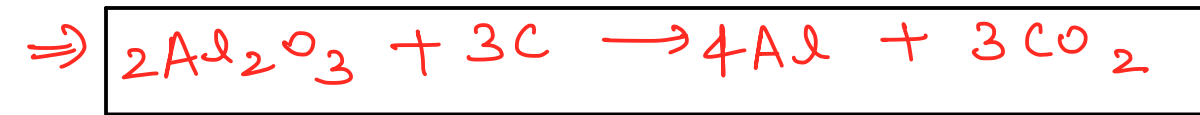
- PURIFIED Al_2O_3 + Na_3AlF_6 or CaF_2 [WHICH LOWERS MELTING POINT AND BRINGS CONDUCTIVITY].

→ CATHODE → STEEL VESSEL WITH LINING OF CARBON

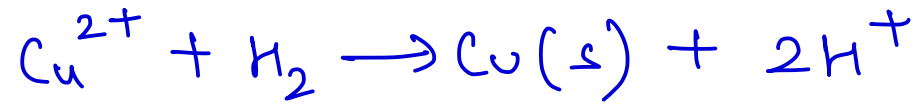
ANODE → GRAPHITE.

FOR EACH Kg OF AL IS PRODUCED, ABOUT 0.5 Kg OF ANODE IS BURNT AWAY.





$\Rightarrow$ COPPER FROM LOW GRADE ORES AND SCRAPS $\Rightarrow$ HYDRO-METALLURGY.



$\rightarrow$ LEACHED OUT USING ACID OR BACTERIA.



3 $\rightarrow$ PURIFICATION OF METAL $\rightarrow$

① DISTILLATION $\rightarrow$ LOW BOILING METALS LIKE ZN AND Hg.

② LIQUATION $\rightarrow$ LOW MELTING METALS LIKE TIN (Sn).

③ ELECTROLYTIC REFINING $\rightarrow$ Zn, Cu.

$\rightarrow$ THE MORE BASIC METAL REMAINS IN SOLUTION AND LESS BASIC ONES GOES TO ANODE MUD.

→ ANODE MUD OF BLISTER COPPER CONTAINS ANTIMONY, SELENIUM, TELLURIUM, GOLD AND PLATINUM.

④ POURING → Cu, Sn, IMPURITY → OXIDE OF METAL.

⑤ ZONE REFINING → IMPURITIES MORE SOLUBLE IN MELT THAN SOLID STATE OF METAL.

- USEFUL FOR PRODUCING SEMI-CONDUCTORS AND OTHER METALS OF VERY HIGH PURITY.

EXAMPLE $\rightarrow$ Ge, Si, B, Ga, In.

⑥ VAPOUR PHASE REFINING —→ MOND'S PROCESS → Ni
—→ VON-ARKELE METHOD → Zr, Ti.

⑦ CHROMATOGRAPHY → MIXTURE IS PUT ONTO A STATIONARY PHASE WHICH MAY BE SOLID OR LIQUID.

⇒ COLUMN CHROMATOGRAPHY → VERY USEFUL FOR

PURIFICATION OF ELEMENTS WHICH ARE AVAILABLE IN MINUTE QUANTITIES AND IMPURITIES ARE NOT VERY DIFFERENT IN CHEMICAL PROPERTIES FROM ELEMENT TO BE PURIFIED.

Al → USED IN EXTRACTION OF CHROMIUM AND MANGANESE FROM OXIDES.

CAST IRON → MOST IMPORTANT FORM OF IRON.